Question			Answer	Marks	AOs		Guidance
15	(a)		$[\mathrm{h}(x)$ **or** $\mathrm{fg}(x) =]\ \sqrt{x^3 - x - 6}$ **oe** $x > 2$	**B1** **B1** **[2]**	**1.1** **1.1**	expression domain	mark the final answer
15	(b)		$\sqrt{18}$ **oe isw FT** *their* $\mathrm{h}(x)$	**B1** **[1]**	**1.1**	allow 4.2426406872… rounded to 2 or more sf	
15	(c)		$\frac{1}{2}\times\frac{3x^2-1}{\sqrt{(x^3-x-6)}}$ or $\frac{3x^2-1}{2\mathrm{h}(x)}$ **oe** *their* $\frac{dh}{dx}$ evaluated at $x = 3$ $\frac{3\sqrt{2}}{13}$ or 0.326356975932 rounded to 2 sf or better	**M1** **A1** **M1** **A1** **[4]**	**3.1a** **1.1** **1.1** **3.2a**	chain rule used all correct	allow one slip in differentiation, eg sign error $\mathrm{h}(x)$ must be correct for first **M1**
			OR $x^2 = y^3 - y - 6 \Longrightarrow 2x\frac{dx}{dy} = 3y^2 - 1$ **oe** $\frac{dx}{dy} = \frac{3y^2-1}{2x}$ *or* $\frac{dy}{dx} = \frac{2x}{3y^2-1}$ substitution of $y = 3$ **and** $x =$ *their* $\sqrt{18}$ $\frac{3\sqrt{2}}{13}$ or 0.326356975932 rounded to 2 sf or better	**M1** **A1** **M1** **A1** **[4]**			allow one slip eg sign error rearrangement to find $\mathrm{h}^{-1}(x)$ explicitly in terms of x followed by differentiation does not score